

Quiz 1

CU2102 CG1103 Theory of Computation

Duration: 1 hour

Maximum Marks: 10

Instructions Provide a proof/counter-example for Question 4. Proof of correctness is not expected for the other questions.

Let $\Sigma = \{a, b\}$. Consider the rational expression e :

$$e \equiv (a^*b^*bbb + a)^*$$

1. Give four words that are **not** in $L(e)$. (1 marks)

Let $K, L \subseteq \Sigma^*$ be two languages, and $u \in \Sigma^*$ be a word. We define $u^{-1}L$ and $K^{-1}L$ as follows:

$$u^{-1}L = \{v \mid uv \in L\}$$

$$K^{-1}L = \bigcup_{u \in K} u^{-1}L$$

2. Give a rational expression for $(aab)^{-1}L(e)$. (1 marks)
3. If L is recognizable, so is $u^{-1}L$. Given a complete DFA for L and a word u , describe how to get a DFA for $u^{-1}L$. (1 marks)
4. Is the following statement true? Substantiate.

$$K \cdot K^{-1}L = L$$

(2 marks)

For $K, L \subseteq \Sigma^*$, define $\text{squash}(L, K)$ as follows:

$$\text{squash}(L, K) = \{uv \mid \exists w \in K, uww \in L\}$$

5. Give a rational expression for $\text{squash}(L(e), L(e))$. (2 marks)
6. If K and L are recognizable, then $\text{squash}(L, K)$ is also recognizable. Given a complete DFA A_L for L and a complete DFA A_K for K , describe how to get an ϵ -NFA for $\text{squash}(L, K)$. (3 marks)